Trends in
Cell Biology

CellPress

Review

What makes genes burst

Varun Sood[1,*] and Tom Misteli[1,*]

Genes burst. Instead of being monotonously transcribed by a steady stream of RNA polymerases, active genes undergo transient and random pulses of transcription that are referred to as gene bursting. This property is ubiquitous and evolutionarily conserved from bacteria to humans, and reflects the inherent stochastic nature of most biological processes. The frequency and duration of gene busting events varies greatly between genes and is now recognized to be controlled by an intricate interplay between transcription factors, chromatin features, and the transcription machinery. Recent findings also point to proximal regulation of bursting by epigenetic chromatin states, a novel role of non-histone modifications, and of distal control of bursting patterns by enhancers. Uncovering the regulatory mechanisms of gene bursting sheds light on how cells maintain a diverse range of gene-specific expression by modulating the different kinetic parameters of bursting.

Highlights

Transcription in most organisms occurs in discontinuous random pulses, referred to as stochastic transcriptional bursting that underlies several physiological roles and pathological effects.

The interplay of transcription factors, promoter nucleosomes, and chromatin remodelers converges on the recruitment of RNA polymerase II for the initiation of transcriptional bursts.

Although post-translational modifications of histone have been associated with different bursting states, causal roles of non-histone modifications in regulating transcriptional bursts have recently been discovered.

Regulation of transcriptional bursting by enhancers can occur either through looping or by enhancer-associated phase-separated clusters containing components of transcriptional machinery and its regulators.

Bursting is a fundamental property of genes

Specialized gene expression programs are at the heart of all biological processes. The faithful execution of any gene program requires the precise **spatio-temporal activation** (see Glossary) and repression of subsets of an organism's genome. A high level of plasticity in regulating the activity of genes is thus a prerequisite to ensure expression of the correct gene sets. Cell- and tissue-specific transcription factors, epigenetic modifications, and the action of regulatory elements, such as enhancers, all contribute to the required high degree of flexibility in gene expression and regulation. In addition, cells have also co-opted the inherent **Brownian motion** of transcription regulators as a dynamic and **stochastic** platform to maximize plasticity in the control of gene expression changes.

Nuclear proteins are inherently dynamic and undergo constrained diffusion in the cell nucleus [1]. Similarly, transcription factors bind largely randomly and transiently to the promoters of target genes [2,3]. The downstream factors along the **gene regulatory axis**, such as transcription cofactors and chromatin remodelers, also reside transiently for only a few seconds at a gene promoter [4,5]. The RNA polymerase II (Pol II) that synthesizes the RNA is then recruited dynamically and associates with the gene promoter where it often pauses before it enters into elongation to generate an RNA molecule [6–8].

Given these dynamic and stochastic properties of the transcription machinery, it is not surprising that transcription itself is also a highly dynamic and heterogeneous event. It is now recognized that the activity of a given gene continuously oscillates between an inactive OFF state and an active ON state (Figure 1A) in a process referred to as stochastic gene bursting. The bursting behavior of a gene is characterized by several properties: The **burst frequency** reflects how often the gene fires in a given period of time (Figure 1B), the **burst size** indicates the total number of RNA polymerases in each burst, the **initiation rate** describes the number of RNA polymerases that initiate transcription per unit time during the burst, and the **burst amplitude** is the peak number of RNA polymerases engaged in transcription (Figure 1B). The ON and OFF states of a gene are

[1]National Cancer Institute, National Institutes of Health (NIH), Bethesda, MD 20896, USA

*Correspondence:
varun.sood@nih.gov (V. Sood) and
mistelit@mail.nih.gov (T. Misteli).

Check for updates

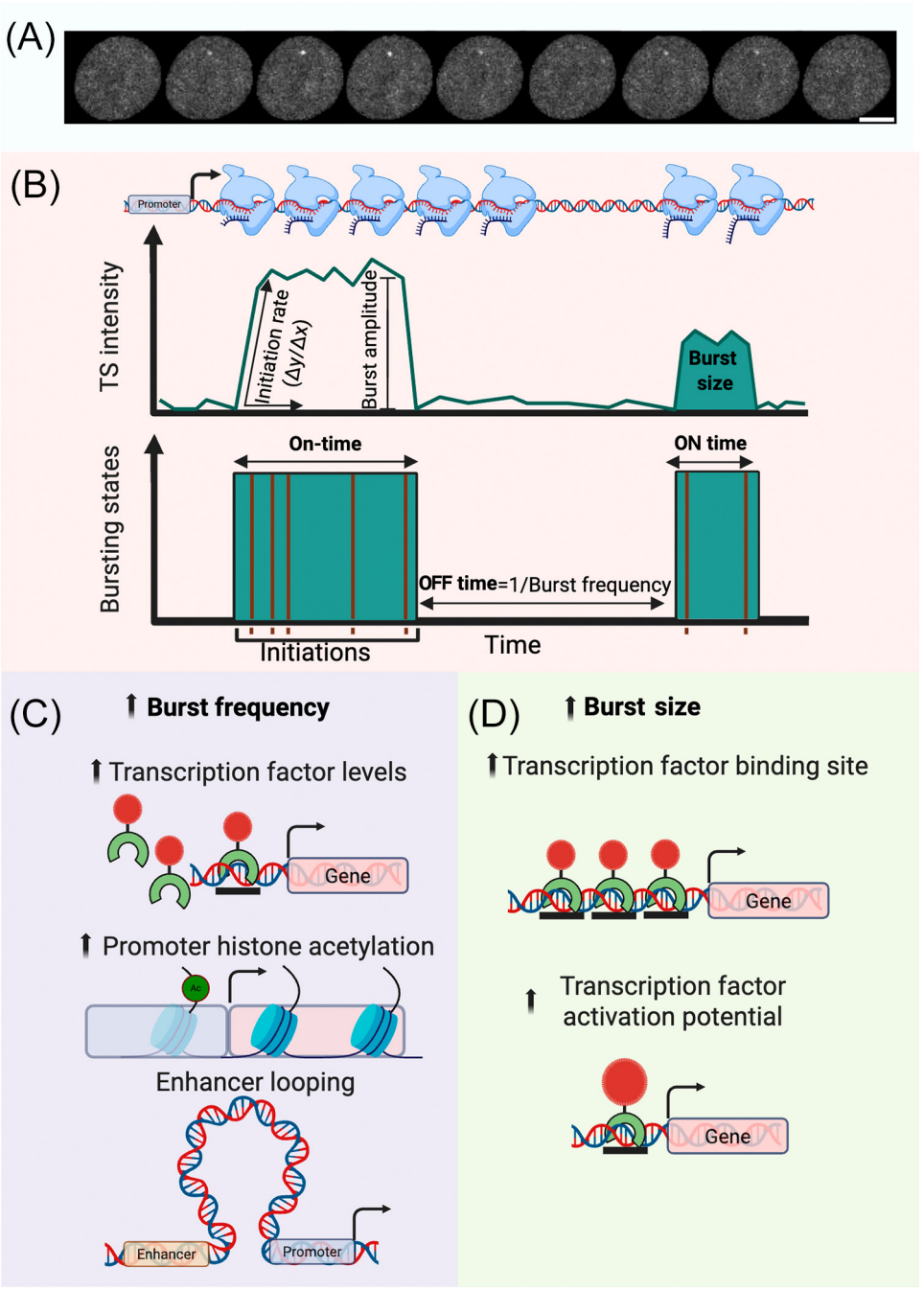

Figure 1. Determinants of transcriptional bursts. The transcription of most active genes is not uniform and instead occurs in random pulses of activity interspersed with varying periods of inactivity, a phenomenon referred to as gene bursting. (A) Representative time-lapse images of two transcriptional bursts of the *ERRFI1* gene in human bronchial epithelial cells represented by bright fluorescent spots in live cells (in frames 2–5 and 7–8). Nascent RNA is visualized using MS2 tagging via the binding of a fluorescent version of the RNA-binding protein of bacteriophage MS2 which recognizes a unique sequence engineered into the *ERRFI1* RNA [25,78]. Scale bar: 5μm. (B) Bursting can be quantitated by image analysis. (Top) Multiple parameters derived from fluoroscence intensity at the transcription site (TS) can be derived from

(Figure legend continued at the bottom of the next page.)

Glossary

+1 and -1 nucleosome: nucleosomes upstream and downstream of a gene's transcription start site.
ATP-dependent chromatin remodelers: enzymes that utilize energy from ATP molecules to reshape chromatin structure.
B-Raf kinase: proto-oncogene involved in cell growth signaling.
Brownian motion: the random thermal motion of biological molecules.
Burst amplitude: indirect measure of the initiation rate represented by the height of the burst signal.
Burst frequency: number of transcriptional bursts in a unit time.
Burst size: number of RNA polymerases in a transcriptional burst.
***Cis*-acting:** part of the same DNA molecule as the gene.
Cognate: a natural interaction partner.
DNA supercoiling: the rotational twists DNA can have beyond the normal relaxed DNA helical structure.
Domains: a functionally or structurally distinct part of a protein.
Dosage compensation: a mechanism to achieve equal gene expression in males and females.
Gene regulatory axis: the chain or network of proteins that regulate gene expression.
H3K4me3: tri-methylation of histone H3 at lysine 4.
Haploinsufficient genes: essential genes that are insufficient for survival when present as single copy.
Histone octamer core: a protein complex consisting of two copies each of four different histone proteins.
Histone tail: the N-terminal region of histone proteins that is unstructured and extensively covalently modified.
Initiation rate: number of RNA polymerases undergoing transcriptional initiation in a unit time.
Intrinsic and extrinsic signals: molecular regulatory cues that originate within or outside of the cells, respectively.
Oct4: transcription factor that contributes to pluripotency.
Off-time: the duration of time between the On-times.
On-time: the duration of time the gene is actively transcribed during a burst.
Sigmoidal function: mathematical function resulting in an S-shaped curve.
Spatio-temporal activation: gene expression in distinct physical locations or at different times.

derived from the best fit to the transcription activity readout, often from single-cell imaging using different kinetic models [9] (Figure 1B). Most commonly a simple two-state model based on gene switching between active and inactive states is used and is generally a good fit [10,11]. However, a single burst is often observed to have multiple closely spaced small bursts [12] or a gene can have multiple OFF states [13], necessitating the need for more complex models [14]. For example, a three-state model in which a gene may fluctuate between an active and inactive state, but may also enter a longer-lasting repressed third state, more accurately reflects its bursting properties in some cases [13].

Stochastic: random.
TATA box: a specific, highly conserved promoter element bound.
Transcriptional noise: the variation in expression levels of genes between individual cells in a population.

Gene bursting has been observed across evolution including in bacteria [15], yeast [16], fly [17], mouse [10], and human cells [18]. Remarkably, the ON/OFF patterns for stochastic bursting are gene-specific and highly variable with individual genes in the same cell exhibiting widely distinct firing patterns [10]. Typically, **ON times** last for a few minutes but can extend to hours for some genes, resulting in bursts ranging from a single to multiple RNA Pol II initiation events [5,19]. **OFF times** on the other hand are often longer, up to several hours, but in some cases can last for days [5,13,19]. Such variations are a major source of single-cell heterogeneity in a cell population and have both physiological roles and pathological effects [20–22].

Although the first evidence of stochastic transcriptional bursting was found almost 50 years ago [23] in the context of the unusual ribosomal RNA genes, the realization that gene bursting is a universal process and insights into the properties and mechanisms of gene bursting have only emerged in the past decade with the advent of single-cell imaging and allele-level studies. This review discusses the current understanding of the mechanistic basis of gene bursting and its regulation.

Stochastic gene bursting is regulated

Although stochastic in nature, gene bursting can be regulated. The net bursting behavior of genes is modulated predominantly by changing the burst frequency or burst size of a gene either individually or in a coordinated fashion across genes for specific biological functions [22,24–26] (Figure 1A,B). For example, **dosage compensation** in *Drosophila melanogaster* males is achieved by increasing the RNA Pol II initiation rate, resulting in higher bursting amplitudes [27], and the spatial expression patterns of gap genes that regulate anterior–posterior patterning in *D. melanogaster* are generated by modulating the fraction of total ON time along the embryonic axis [26,28]. On the other hand, synchronous bursting of key genes can confer chemoresistance in melanoma cells exposed to the **B-RAF kinase** inhibitor vemurafenib by propagating downstream changes in signaling and gene expression profiles [22,24]. Bursting parameters can also act as levers to fine-tune both expression levels and variability. For instance, the expression of genes amplified by shortening the OFF time leads to less variability within the population as opposed to increasing the burst size [29]. Increased variability can be utilized for adaptation to environmental stress and generally correlates with increased fitness [30,31]. As an example, a subset of yeast cells with higher expression of the gene involved in trehalose synthesis, *TSL1*, grow slower under normal conditions compared to the rest of the population but exhibit better fitness upon heat stress [30]. Gene expression heterogeneity is also key to maintaining

model fitting of experimental data. Key parameters TS intensity are: initiation rate defined as the number of RNA polymerases per unit time; burst amplitude, reflecting the maximum number of RNA polymerases detected; and burst size, the number of RNA polymerases in a burst represented by the area under the curve. The ON time represents the duration of the bursts and the OFF time represents the time between bursts. (C,D) Schematic of different factors that preferentially impact on either burst frequency or burst size by decreasing the OFF time (C) or by increasing the ON time (D), respectively. (C) Transcription factor levels, epigenetic histone acetylation marks, and the proximity of enhancers all affect transcription factor recruitment to the promoter and primarily affect burst frequency, whereas (D) the strength of the transcription factor binding events and the ability of a transcription factor to activate and sustain transcription primarily determine burst size.

developmental plasticity and differentiation, as illustrated in mouse embryonic stem cells (mESCs) where cells with high levels of the **OCT4** transcription factor arising from spontaneous gene expression fluctuations create high accessibility at differentiation enhancers and thus promote differentiation towards neuroectodermal and mesendodermal fates [32]. Similarly, mouse hematopoietic cells with spontaneously arising high and low levels of the stem cell marker SCA1 show marked differentiation preferences between myeloid versus erythroid lineages, respectively [21]. Interestingly, the expression heterogeneity of differentiation genes often spikes immediately before cell fate decisions during development in *Dictyostelium discoideum* and mouse [33,34], further pointing to a functional role of bursting dynamics in developmental processes. However, **transcriptional noise** can also be deleterious – such as the heterogeneity of circadian-related genes which is associated with aging [35]. Furthermore, heterogeneity in the expression of dosage-sensitive genes in yeast, including essential and **haploinsufficient genes**, can be detrimental for growth [36]. These select observations illustrate that stochastic bursting is an emergent property that has regulatory consequences for a wide range of biological functions.

Some bursting features are genetically coded in the promoter sequences of a gene. This is well illustrated in the duplicated copies of the *actin* gene in *D. discoideum* which show widely different bursting rates because of variation in their promoter sequences [37], and switching promoter sequences recapitulates the differences in burst size, burst frequency, and duration between the gene copies [37]. Such variations can arise from differences in the number and/or strength of transcription factor binding sites or core promoter elements. For example, in yeast a weak Gal4 binding site reduces burst size by decreasing the dwell time of the protein on DNA [38], but the presence of a **TATA box** promoter element can increase both the burst size and duration by promoting longer active states and higher initiation rates [39–41]. This leads to a rapid and more stochastic transcriptional profile for stress-induced genes, which facilitates fitness under stress [42,43]. In addition, other core promoter elements at the transcription start-site regulate the basal burst frequency and drive stochastic bursting synergistically with the TATA box [39–41,44] and associated enhancers [45]. Conversely, the upstream core promoter CCAAT element that recruits nuclear factor Y regulates both the frequency and size of bursting upon induction [44]. Thus, the interplay between the *cis*regulatory elements and the random dynamics of *trans* factors influences the kinetic parameters of stochastic bursting.

Proximal regulation of bursting patterns by chromatin

Multiple gene-regulatory events, including transcription factor binding, nucleosome occupancy, and the action of chromatin remodelers and epigenetic states mediated by histone modifications, shape the frequency and duration of gene bursts (Figure 1B). Although these regulators often preferentially alter a particular bursting parameter (Figure 1C,D), the observed effects are rarely limited to a single bursting feature, and this may reflect biological properties and assay-specific effects [14,46].

Transcription factors and nucleosomes

The probability of bursting is highly dependent on the kinetics of transcription factor binding to chromatin [47]. Both the concentration and binding affinity of transcription factors have prominent and overlapping effects on bursting kinetics. A higher nuclear concentration of a transcription factor increases its bound fraction, which in most cases increases the burst frequency [13,38,48,49] (Figure 1C). For example, higher levels of Gal4 in yeast and p-ERK in human cells increase the burst frequency of *GAL* and *c-FOS*, respectively [38,49]. By contrast, higher levels of some transcription factors, for example Myc in human cells, may increase the burst size [50,51]. On the DNA side, the sequence of transcription factor binding sites affects their binding affinity and

consequently the transcription factor dwell time, which mostly regulates the burst duration [38,49,52] or burst size [26,53] (Figure 1D), but in some cases can also exclusively regulate burst frequency [54]. Overall, despite these variations between different genes, both transcription factor dwell time and concentration correlate positively with bursting, although transcription factor dwell times usually have a bigger impact than concentration [54]. Of note is the fact that transcription factors dwell times are typically on the order of seconds, and are thus much shorter than the duration of the ON or OFF periods. This behavior is well in line with the finding that persistent steady-states are typically generated by recurrent dynamic binding events of individual proteins [2,55,56].

Transcription factors interact *in vivo* with promoter DNA in the context of chromatin. Nucleosomes, the basic units of chromatin organization consist of a **histone octamer core** that substantially decreases transcription factor binding by restricting access to target sites [38]. A high density of nucleosomes is typically associated with reduced transcription factor dwell time on chromatin [38,57] and increased dissociation rates [57]. The resulting lower transcription factor occupancy has been linked to reduced bursting. In support, higher transcriptional noise has been reported at higher nucleosome-occupancy loci for reporter genes [58] and upon activation of latent HIV with NF-κB [59]. Furthermore, the incorporation of nucleosome-disfavoring sequences into the promoter region decreases transcriptional noise by increasing the probability of bursting [29]. In agreement, the turnover rates of nucleosomes on chromatin correlate with gene expression levels and are usually higher at promoters than in the gene body [60]. In addition to nucleosomes, **DNA supercoiling** also contributes to bursting control by restricting transcription factor binding and impeding transcription initiation [61].

The dynamic and stochastic variation of nucleosome occupancy can also indirectly affect both the expression/noise balance as well as transcription factor specificity [29]. Expression levels and transcriptional noise are usually anti-correlated, but small decreases in promoter nucleosome occupancy can decrease noise without changing expression, thus partially decoupling the two [29]. In addition, the stochastic removal and re-formation of promoter nucleosomes can also add a kinetic proof-reading step for transcription factor binding [29]. This process creates a delay in transcription activation which may contribute to longer burst durations relative to the rapid binding of transcription factors [55,56] and could add an energy-dissipating step that favors specific binding. The latter effect is significant because the dwell time differences between specific and non-specific binding of most transcription factors are relatively small compared to the burst durations [62] and because non-**cognate** transcription factors usually outnumber specific transcription factors at target promoters [63]. In a similar fashion, nucleosomes can promote cooperativity and specificity between multiple transcription factors that bind to different sites on a nucleosome [47]. The dynamic and stochastic nature of the nucleosome thus offers both an effective quantitative and qualitative control of bursting.

Chromatin remodelers

Multiple classes of **ATP-dependent chromatin remodelers** impact bursting by shaping the nucleosome architecture [64]. Although the underlying DNA sequences can have different propensities to wrap around the nucleosome [65], the overlapping action of multiple remodelers may generate energetically unfavorable chromatin architectures, such as nucleosome-depleted regions, that facilitate bursting in a sequence-independent fashion [66]. Although the various classes of remodelers differ in their effect on nucleosomal occupancy and relative positioning as a result of differences in **domains** and subunits across different multi-subunit remodeling complex, all remodelers share a conserved ATPase motor that utilizes DNA translocation as the fundamental mechanism to restructure DNA–histone contacts within nucleosomes. Consistent with their role in modulating nucleosome occupancy, early deletion studies in yeast targeting the SWI/

SNF (Snf6) or INO80 (Arp8) remodelers showed consistent increases in transcriptional noise [67]. More recent single-molecule studies indicate that most remodelers bind only transiently, and often repeatedly, to the same chromatin locus, which leads to stochastic variation in their accessibility for the transcription initiation machinery [68]. Studies in yeast have further highlighted that different classes of remodelers also act cooperatively to regulate different aspects of bursting. For example, both RSC and SWI/SNF remodelers have partly redundant impacts on both nucleosome architecture and bursting at the *GAL10* promoter [68]. In support, depletion of RSC (Sth1) decreased the occupancy of the partially unwrapped promoter nucleosome at the *GAL* upstream activating sequence (UAS_{GAL}), but not of the stable promoter nucleosomes, and increased both the initiation time and the OFF time [68]. Depletion of SWI/SNF (Swi2), on the other hand, increased nucleosome occupancy at the promoter and increased only the initiation time [68]. However, both remodelers were necessary to maintain the nucleosome-free region between the **+1 and −1 nucleosomes**, and double depletion showed a partially additive phenotype [68]. These effects of remodeling on bursting result from the crosstalk with transcription factor binding at promoters. For example, RSC and Gal4 mutually increase their binding at the UAS_{GAL} of the *GAL10* promoter that additively affects the initiation rate and cooperatively increases the burst frequency [68]. A similar relationship with RSC was observed with TBP at the *GAL10* TATA box [68] and with the Ace1 transcription factor at the *CUP1* promoter [52]. Taken together, the stochastic and overlapping actions of multiple remodelers facilitate transcription factor binding and in this way contribute to regulating gene bursting.

Epigenetic modifications

The octamer core of nucleosomes is extensively modified at **histone tails**, and distinct sets of epigenetic modifications are associated with different expression states [69–71]. Histone tail modifications have also been implicated in gene bursting. These modifications may affect gene activation either directly or indirectly by recruiting transcription cofactors with corresponding recognition domains, or they may act by changing nucleosome stability and/or promote pause-release [69–71], both of which affect bursting [72–74]. Indeed, several comparative studies have linked epigenetic states to bursting. For example, promoter epigenetic marks such as histone H3 lysine 4 trimethylation (**H3K4me3**) from yeast to humans correlate genome-wide with both burst size and frequency, whereas gene body marks such as H3K79me2 associate strongly with only burst frequency [75]. Promoter association with H3K4me4 is also required for the inheritance of bursting patters through cell lineages for multiple genes in *D. discoideum* [76]. In addition, changes in burst size across several genes and cell lines are observed upon a global increase in acetylation [10,59,77]. Although global acetylation changes include both histone and non-histone targets that may impact bursting [78], site-specific elevation in promoter acetylation increases burst frequency, for example in the circadian gene *Bmal1* in mouse NIH3T3 cells [79] (Figure 1C). In agreement, promoter recruitment of the p300 acetyltransferase increases chromatin accessibility locally, resulting in a less noisy expression pattern for multiple genes in human cells even when their expression levels are unchanged [80]. However, the relative roles of other histone modifications and how pervasive these effects are remains unclear.

Revisiting the role of epigenetic states in gene bursting

Although various observations point to a role of histone modifications in gene bursting, the extent of their contribution has recently come under closer scrutiny. Specifically, several studies suggest that the impact of epigenetic modifications is relatively modest and is restricted to a subset of genes in particular cell lines [10,81,82] (Figure 2A).

Several lines of evidence question the assumed strong causal role of epigenetic modifications, especially of promoter nucleosomes, in bursting. In particular, global increases in protein acetylation

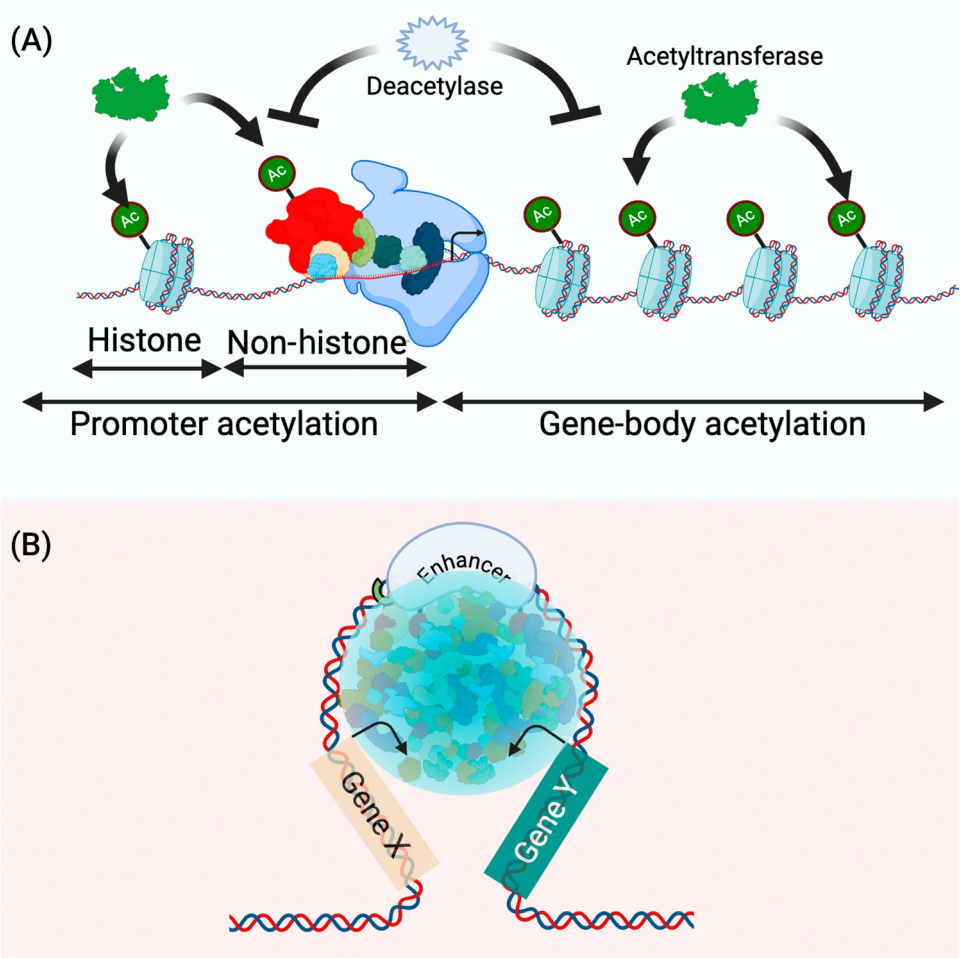

Figure 2. New regulators of transcriptional bursting. Several new aspects of gene bursting have recently been uncovered. (A) Although acetylation (Ac) of histones at the promoter (left) and in the gene body (right) has long been associated with determining bursting patterns, recent observations have documented a prominent role for promoter-associated non-histone protein acetylation (middle), especially of RNA polymerase-interacting proteins (red) such as the Integrator complex, in regulating bursting. The acetylation status of histone and non-histone proteins is determined by multiple acetyltransferases and deacetylases that have overlapping target specificities. (B) Enhancer-associated liquid–liquid phase-separated condensates promote busting of targets genes. The proposed mechanism for this effect is the recruitment of transcription factors (blue, green) into large (200–1000 nm) condensates (pale blue) which effectively increase the local concentration of transcription factors and in this way alter burst patterns of genes associated with the condensate. Of note is that in this model there is no requirement for the enhancer to directly contact the promoter regions of the target genes.

changes do not appear to affect bursting globally [10,83]. More tellingly, genome-wide comparison of promoter histone H3 lysine 27 acetylation (H3K27ac) and bursting revealed only a weak correlation [81]. Bursting rates and promoter acetylation have also been found to be stoichiometrically uncoupled for several genes, both in the native condition and upon global hyperacetylation, perhaps most tellingly upon tethering of acetylation modifiers near to the promoter region of target genes [78].

One of the reasons to revisit the contribution of epigenetic modifications to bursting is the fact that some effects previously attributed to histone modifications, such as those induced by

acetyltransferase inhibitors, may in fact not be due to their modification of histones [78,84–86]. Evidence for a prominent non-histone-mediated role for protein acetylation in gene bursting comes from a recent screening approach [78]. When testing the effect of a library of 140 epigenetic modulators on the bursting of 14 native genes across multiple cell lines [78] in a high-throughput fluorescence *in situ* hybridization single-cell assay, the impact of the epigenetic modulators on the ON and OFF times was highly gene- and cell line-specific [78]. Among a wide range of post-translational modifications, changes in acetylation had the strongest impact, and led to a universal decrease in OFF times but gene-specific changes in ON times. However, instead of being mediated by promoter histone acetylation, the observed effects were due to acetylation of N-terminal lysines on the Integrator subunit 4 (Ints4) – a subunit of the Integrator complex that controls the pause-release of RNA Pol II by cleavage of the 5′-end of the nascent RNA [87,88] (Figure 2A). Acetylation of Ints4 leads to a decrease in target gene bursting by reducing its interaction with multiple factors involved in RNA Pol II initiation, elongation, and mRNA splicing [78]. The identification of Integrator as a modulator of acetylation is likely the proverbial tip of the iceberg and there are probably other regulatory pathways that act via acetylation or other post-translational modifications. For example, a similar impact of acetylation has been reported for gene activation by p53 [84] and SOX2 [85]. The details of the interplay and relative contributions of promoter histone modifications and non-histone modifications remain to be elucidated (Figure 2A).

The model that is emerging from the combined observations of various effectors of bursting is that gene bursting is primarily controlled at the level of RNA Pol II and that the effect of modulators of bursting, such as transcription factors, nucleosome density, and epigenetic modifications, converge on RNA Pol II (Figure 2A). This mechanism offers multiple points of control, particularly at the level of RNA Pol II recruitment, initiation, and pause-release. Regarding recruitment, the probability to nucleate gene bursts is a function of both *cis* and *trans* factor-mediated recruitment of RNA Pol II to the promoter [38,47]. RNA Pol II recruitment in turn is affected by the dwell time of transcription factors, which can be regulated, for example via post-translational modification [84,89]. Furthermore, at promoters with multiple transcription factor binding sites, the individual molecules of a transcription factor may cooperate during active bursts and promote longer bursts by successive binding of multiple transcription factor molecules [47]. Similarly, multiple RNA Pol II may initiate while the transcription factor remains bound at the promoter, whereas transcription factor dissociation conversely leads to termination of the burst [38,47]. A further point of burst control is the initiation and pause-release of RNA Pol II that can occur without changing the recruitment rates in response to multiple transcriptional perturbations [72]. The initiation rates during the burst are affected by the strength of the transactivation domain of the transcription factors involved [49]. They are also correlated positively with burst size and determine the amplitude of the burst [49] (Figure 1D). These observations make it clear that the combination of recruitment, initiation rate, and RNA Pol II pause-release determine overall bursting patterns.

Distal regulators of bursting patterns

Enhancers are key regulators of metazoan gene expression that determine cell identity and response to **intrinsic and extrinsic signals** [39,90]. Enhancers are **cis-acting** elements that promote the transcription of target genes, typically 10 to several hundred kb away from them [91], and are characterized by specific histone acetylation patterns; they often also comprise clusters of sequence-specific transcription factors and coactivators that act as gene regulatory hubs [92–94]. Recent observations are beginning to link enhancer function to gene bursting.

One example for enhancer-mediated bursting regulation is the behavior of segmentation genes in *D. melanogaster*. These genes are either proximal or distal to a shared enhancer, and their

looping into an enhancer hub via tethering elements is required for spatiotemporal co-bursting [95]. Furthermore, sustained proximity to a distal even-skipped enhancer is necessary for transcriptional activation of individual loci during development [96]. In comparative genome-wide bursting studies in human cells, enhancers primarily modulate burst frequency [39] and their effect correlates with the strength of the enhancer [97] (Figure 1C). Enhancer deletion [98] or swapping [99] confirm the functional roles of enhancers in bursting. The effect of enhancers on bursting decreases rapidly with a **sigmoidal function** over increasing distances [97] and is constrained by the boundaries of chromatin domains and binding sites for the CTCF architectural chromatin protein, as shown for *Sox9* in human embryonic stem cells [100].

Enhancers are characterized by high levels of H3K27ac and often nucleate a non-coding transcript that can impact bursting [86]. In genome-wide studies, enhancer acetylation levels correlate positively with bursting of the cognate gene [39]. Furthermore, site-specific changes in enhancer acetylation affect bursting [101]; for example, the acetylation of an enhancer of *FOS* increased burst duration, whereas decreasing the acetylation increased the OFF time in mouse neurons [101]. In agreement with a role of non-histone acetylation in bursting, enhancers also act as a source of acetylated transcription factors that can encounter a promoter in a less stringent distance-dependent manner [86,102]. A high level of histone acetylation and transcription-associated factors can initiate non-coding transcripts [103] that can affect the bursting of the cognate gene in *cis*, and enhancer transcripts that initiate upstream of transcription factor binding sites within enhancers lead to a decrease in bursting through disruption of transcriptional hubs at cognate gene loci [104]. Thus, the interplay between enhancer acetylation and transcription activity can dictate the effectiveness of gene bursting.

Although traditional models suggest that enhancers physically interact with their target genes [96,98], the prevailing view of how enhancers function envisages that they act via the formation of transcriptional condensates [53,92,93,105–108] (Figure 2B). In this model, enhancers nucleate the formation of membraneless protein assemblies formed by low-affinity interactions between proteins containing intrinsically disordered regions [53,93], which are prevalent in many transcription factors, coactivators, and RNA Pol II [53,93]. Because condensates are typically on the order of several hundred nanometers in size, direct physical interaction between enhancer and target gene is not a general necessity in these models [109,110]. Indeed, the proximity of an enhancer to the target gene is not a requirement for its effect on gene bursting [53,93,105]. A recent live-cell study showed that transcription-dependent Pol II condensates upregulate both the burst size and the fraction of *Sox2* genes that are expressed in mouse embryonic stem cells at distances typically in the range of hundreds of nanometers, but up to 1 μm [105]. Upregulation correlated better with the distance between the gene and the condensate rather than its distance from the enhancer [105]. Both the enhancer and the chromatin-looping factor cohesin were necessary for increased bursting, and a model has been suggested in which the target gene interacts transiently with the condensate and with its regulatory element to control bursting [105]. Furthermore, a single enhancer can simultaneously activate co-bursting of two reporter genes, thereby precluding the need for gene-specific looping [99]. To further understand the temporal and causal relationship between transcription factor condensates and bursting, a detailed investigation was recently carried out using synthetic enhancers in *D. melanogaster* [53]. These enhancers attract high concentrations of transcriptional activators which form a condensate via their intrinsically disordered regions. Interestingly, the accumulation of these factors temporally preceded the gene bursts, which in turn correlated directly with condensate size [53]. These findings show that enhancers affect gene bursting via distal regulatory events.

Concluding remarks

Gene bursting has only relatively recently been recognized as a fundamental feature of most genes. Studies over the past few years have started to delineate some of the regulatory mechanisms of bursting and have identified some generic chromatin features such as nucleosome density and histone modifications that are modulators of bursting. More in-depth analysis of the complex interplay of chromatin structure, epigenetic state, and transcription factor binding will be necessary to fully characterize the contribution of each of these chromatin features to bursting control. In addition, a major role for bursting control via post-translational acetylation of an RNA Pol II interacting complex has emerged. This finding has multiple important implications. First, it points to the existence of a prominent RNA Pol II-mediated checkpoint of bursting at the level of transcriptional initiation. Second, the involvement of acetylation as a regulatory mechanism suggests the presence of upstream regulatory events that most likely sense and transmit intrinsic or extrinsic cellular stimuli. Beyond fully elucidating the mechanistic basis for the role of Integrator in bursting, it stands to reason that additional transcriptional modulators of bursting remain to be discovered, many of which probably act in cell type- or tissue-specific contexts. An obvious question for future studies concerns the identity of the upstream signaling pathways that feed into bursting regulators such as Integrator.

The characterization of the phenomenon of gene bursting and its mechanistic dissection have lagged behind that of transcriptional gene regulation. One of the main reasons for this delay is that probing of bursting requires single-cell methodologies rather than the bulk approaches traditionally used for gene expression studies. With the rapid development of an increasing arsenal of single-cell genomic, transcriptomic, and biochemical methods combined with the single-cell capabilities of advanced imaging, it will now be possible to rapidly make progress in uncovering the mechanisms of gene bursting. Future studies should focus on the comprehensive identification of regulatory genome elements such as enhancers and repressors, as well as on cataloguing the protein effectors of bursting, possibly via probing the RNA Pol II interactome. Furthermore, the use of synthetic biology approaches to reproduce bursting behavior will be the ultimate test of our mechanistic understanding of gene bursting. These experimental studies should also be complemented by computational simulations which can probe bursting behavior *in silico* and generate predictions for experimental validation. These approaches will shed much needed insight into one of the least-understood, ubiquitous, fundamental, and evolutionarily highly conserved phenomena in cell biology (see Outstanding questions).

Outstanding questions

Given that only a small fraction of transcription factor binding events lead to bursting, what determines whether the binding of a transcription factor will trigger a transcriptional burst?

What are the upstream signaling pathways that affect bursting patterns in physiological and pathological settings?

How broad is the scope of non-histone acetylation and other post-translational modifications in regulating transcriptional bursts, and what are their mechanisms of action?

What factors determine the enhancer specificities and their mode of regulating cognate gene bursting, for example through looping or transcription-dependent condensates?

Acknowledgments

We dedicate this review to the memory of Dr Michael Bustin who pioneered the exploration of chromatin proteins and was an inspiring collaborator, generous colleague and supportive mentor to T.M. We thank members of the laboratory of T.M. for their feedback. This work was supported by the Intramural Research Program of the NIH, National Cancer Institute, Center for Cancer Research grant ZIABC010309-25 to T.M. The contributions of the NIH authors were made as part of their official duties as NIH federal employees, are in compliance with agency policy requirements, and are considered Works of the United States Government. However, the findings and conclusions presented in this paper are those of the author(s) and do not necessarily reflect the views of the NIH or the US Department of Health and Human Services.

Declaration of interests

The authors declare no competing interests.

References

1. Misteli, T. (2001) Protein dynamics: implications for nuclear architecture and gene expression. *Science* 291, 843–847
2. Hager, G.L. *et al.* (2009) Transcription dynamics. *Mol. Cell* 35, 741–753
3. Lu, F. and Lionnet, T. (2021) Transcription factor dynamics. *Cold Spring Harb. Perspect. Biol.* 13, a040949
4. Kim, J.M. *et al.* (2021) Single-molecule imaging of chromatin remodelers reveals role of ATPase in promoting fast kinetics of target search and dissociation from chromatin. *eLife* 10, e69387
5. Meeussen, J.V.W. and Lenstra, T.L. (2024) Time will tell: comparing timescales to gain insight into transcriptional bursting. *Trends Genet.* 40, 160–174

6. Baek, I. *et al.* (2021) Single-molecule studies reveal branched pathways for activator-dependent assembly of RNA polymerase II pre-initiation complexes. *Mol. Cell* 81, 3576–3588
7. Core, L. and Adelman, K. (2019) Promoter-proximal pausing of RNA polymerase II: a nexus of gene regulation. *Genes Dev.* 33, 960–982
8. Nguyen, V.Q. *et al.* (2021) Spatiotemporal coordination of transcription preinitiation complex assembly in live cells. *Mol. Cell* 81, 3560–3575
9. Tunnacliffe, E. and Chubb, J.R. (2020) What is a transcriptional burst? *Trends Genet.* 36, 288–297
10. Suter, D.M. *et al.* (2011) Mammalian genes are transcribed with widely different bursting kinetics. *Science* 332, 472–474
11. Peccoud, J. and Ycart, B. (1995) Markovian modeling of gene-product synthesis. *Theor. Popul. Biol.* 48, 222–234
12. Tantale, K. *et al.* (2016) A single-molecule view of transcription reveals convoys of RNA polymerases and multi-scale bursting. *Nat. Commun.* 7, 12248
13. Rodriguez, J. *et al.* (2019) Intrinsic dynamics of a human gene reveal the basis of expression heterogeneity. *Cell* 176, 213–226
14. Zhang, Q. *et al.* (2024) Transcriptional bursting dynamics in gene expression. *Front. Genet.* 15, 1451461
15. Golding, I. *et al.* (2005) Real-time kinetics of gene activity in individual bacteria. *Cell* 123, 1025–1036
16. Zenklusen, D. *et al.* (2008) Single-RNA counting reveals alternative modes of gene expression in yeast. *Nat. Struct. Mol. Biol.* 15, 1263–1271
17. Pare, A. *et al.* (2009) Visualization of individual Scr mRNAs during *Drosophila* embryogenesis yields evidence for transcriptional bursting. *Curr. Biol.* 19, 2037–2042
18. Raj, A. *et al.* (2006) Stochastic mRNA synthesis in mammalian cells. *PLoS Biol.* 4, e309
19. Dar, R.D. *et al.* (2012) Transcriptional burst frequency and burst size are equally modulated across the human genome. *Proc. Natl. Acad. Sci. U. S. A.* 109, 17454–17459
20. Abranches, E. *et al.* (2014) Stochastic NANOG fluctuations allow mouse embryonic stem cells to explore pluripotency. *Development* 141, 2770–2779
21. Chang, H.H. *et al.* (2008) Transcriptome-wide noise controls lineage choice in mammalian progenitor cells. *Nature* 453, 544–547
22. Shaffer, S.M. *et al.* (2017) Rare cell variability and drug-induced reprogramming as a mode of cancer drug resistance. *Nature* 546, 431–435
23. McKnight, S.L. and Miller, O.L., Jr. (1979) Post-replicative nonribosomal transcription units in *D. melanogaster* embryos. *Cell* 17, 551–563
24. Torre, E.A. *et al.* (2021) Genetic screening for single-cell variability modulators driving therapy resistance. *Nat. Genet.* 53, 76–85
25. Wan, Y. *et al.* (2021) Dynamic imaging of nascent RNA reveals general principles of transcription dynamics and stochastic splice site selection. *Cell* 184, 2878–2895
26. Zoller, B. *et al.* (2018) Diverse spatial expression patterns emerge from unified kinetics of transcriptional bursting. *Cell* 175, 835–847
27. Forbes Beadle, L. *et al.* (2023) Modulation of transcription burst amplitude underpins dosage compensation in the *Drosophila* embryo. *Cell Rep.* 42, 112382
28. Fukaya, T. (2021) Dynamic regulation of anterior-posterior patterning genes in living *Drosophila* embryos. *Curr. Biol.* 31, 2227–2236
29. Dadiani, M. *et al.* (2013) Two DNA-encoded strategies for increasing expression with opposing effects on promoter dynamics and transcriptional noise. *Genome Res.* 23, 966–976
30. Carey, J.N. *et al.* (2018) Regulated stochasticity in a bacterial signaling network permits tolerance to a rapid environmental change. *Cell* 173, 196–207
31. Yang, X. *et al.* (2022) Silent transcription intervals and translational bursting lead to diverse phenotypic switching. *Phys. Chem. Chem. Phys.* 24, 26600–26608
32. Strebinger, D. *et al.* (2019) Endogenous fluctuations of OCT4 and SOX2 bias pluripotent cell fate decisions. *Mol. Syst. Biol.* 15, e9002
33. Antolovic, V. *et al.* (2017) Generation of single-cell transcript variability by repression. *Curr. Biol.* 27, 1811–1817
34. Goolam, M. *et al.* (2016) Heterogeneity in Oct4 and Sox2 targets biases cell fate in 4-cell mouse embryos. *Cell* 165, 61–74
35. Wolff, C.A. *et al.* (2023) Defining the age-dependent and tissue-specific circadian transcriptome in male mice. *Cell Rep.* 42, 111982
36. Lehner, B. (2008) Selection to minimise noise in living systems and its implications for the evolution of gene expression. *Mol. Syst. Biol.* 4, 170
37. Tunnacliffe, E. *et al.* (2018) Promoter-mediated diversification of transcriptional bursting dynamics following gene duplication. *Proc. Natl. Acad. Sci. U. S. A.* 115, 8364–8369
38. Donovan, B.T. *et al.* (2019) Live-cell imaging reveals the interplay between transcription factors, nucleosomes, and bursting. *EMBO J.* 38, e100809
39. Larsson, A.J.M. *et al.* (2019) Genomic encoding of transcriptional burst kinetics. *Nature* 565, 251–254
40. Pimmett, V.L. *et al.* (2021) Quantitative imaging of transcription in living *Drosophila* embryos reveals the impact of core promoter motifs on promoter state dynamics. *Nat. Commun.* 12, 4504
41. Yokoshi, M. *et al.* (2022) Dynamic modulation of enhancer responsiveness by core promoter elements in living *Drosophila* embryos. *Nucleic Acids Res.* 50, 92–107
42. Blake, W.J. *et al.* (2006) Phenotypic consequences of promoter-mediated transcriptional noise. *Mol. Cell* 24, 853–865
43. Huisinga, K.L. and Pugh, B.F. (2004) A genome-wide housekeeping role for TFIID and a highly regulated stress-related role for SAGA in *Saccharomyces cerevisiae*. *Mol. Cell* 13, 573–585
44. Hendy, O. *et al.* (2017) Differential context-specific impact of individual core promoter elements on transcriptional dynamics. *Mol. Biol. Cell* 28, 3360–3370
45. Li, L. *et al.* (2021) Two promoters integrate multiple enhancer inputs to drive wild-type knirps expression in the *Drosophila melanogaster* embryo. *Genetics* 219, iyab154
46. Rodriguez, J. and Larson, D.R. (2020) Transcription in living cells: molecular mechanisms of bursting. *Annu. Rev. Biochem.* 89, 189–212
47. Pomp, W. *et al.* (2024) Transcription factor exchange enables prolonged transcriptional bursts. *Mol. Cell* 84, 1036–1048
48. Lee, R.E. *et al.* (2014) Fold change of nuclear NF-kappaB determines TNF-induced transcription in single cells. *Mol. Cell* 53, 867–879
49. Senecal, A. *et al.* (2014) Transcription factors modulate c-Fos transcriptional bursts. *Cell Rep.* 8, 75–83
50. Patange, S. *et al.* (2022) MYC amplifies gene expression through global changes in transcription factor dynamics. *Cell Rep.* 38, 110292
51. Trzaskoma, P. *et al.* (2024) 3D chromatin architecture, BRD4, and Mediator have distinct roles in regulating genome-wide transcriptional bursting and gene network. *Sci. Adv.* 10, eadl4893
52. Mehta, G.D. *et al.* (2018) Single-molecule analysis reveals linked cycles of RSC chromatin remodeling and Ace1p transcription factor binding in yeast. *Mol. Cell* 72, 875–887
53. Kawasaki, K. and Fukaya, T. (2023) Functional coordination between transcription factor clustering and gene activity. *Mol. Cell* 83, 1605–1622
54. Popp, A.P. *et al.* (2021) Altering transcription factor binding reveals comprehensive transcriptional kinetics of a basic gene. *Nucleic Acids Res.* 49, 6249–6266
55. Lammers, N.C. *et al.* (2020) A matter of time: using dynamics and theory to uncover mechanisms of transcriptional bursting. *Curr. Opin. Cell Biol.* 67, 147–157
56. Veitia, R.A. (2025) Rethinking transcription factor dynamics and transcription regulation in eukaryotes. *Trends Biochem. Sci.* 50, 376–384
57. Luo, Y. *et al.* (2014) Nucleosomes accelerate transcription factor dissociation. *Nucleic Acids Res.* 42, 3017–3027
58. Dey, S.S. *et al.* (2015) Orthogonal control of expression mean and variance by epigenetic features at different genomic loci. *Mol. Syst. Biol.* 11, 806
59. Wong, V.C. *et al.* (2018) NF-kappaB–chromatin interactions drive diverse phenotypes by modulating transcriptional noise. *Cell Rep.* 22, 585–599

60. Dion, M.F. *et al.* (2007) Dynamics of replication-independent histone turnover in budding yeast. *Science* 315, 1405–1408
61. Patel, H.P. *et al.* (2023) DNA supercoiling restricts the transcriptional bursting of neighboring eukaryotic genes. *Mol. Cell* 83, 1573–1587
62. Suter, D.M. (2020) Transcription factors and DNA play hide and seek. *Trends Cell Biol.* 30, 491–500
63. Samee, M.A.H. (2023) Noncanonical binding of transcription factors: time to revisit specificity? *Mol. Biol. Cell.* 34, pe4
64. Rando, O.J. and Winston, F. (2012) Chromatin and transcription in yeast. *Genetics* 190, 351–387
65. Segal, E. *et al.* (2006) A genomic code for nucleosome positioning. *Nature* 442, 772–778
66. Yen, K. *et al.* (2012) Genome-wide nucleosome specificity and directionality of chromatin remodelers. *Cell* 149, 1461–1473
67. Raser, J.M. and O'Shea, E.K. (2004) Control of stochasticity in eukaryotic gene expression. *Science* 304, 1811–1814
68. Brouwer, I. *et al.* (2023) Dynamic epistasis analysis reveals how chromatin remodeling regulates transcriptional bursting. *Nat. Struct. Mol. Biol.* 30, 692–702
69. Policarpi, C. *et al.* (2024) Systematic epigenome editing captures the context-dependent instructive function of chromatin modifications. *Nat. Genet.* 56, 1168–1180
70. Slaughter, M.J. *et al.* (2021) HDAC inhibition results in widespread alteration of the histone acetylation landscape and BRD4 targeting to gene bodies. *Cell Rep.* 34, 108638
71. Bannister, A.J. and Kouzarides, T. (2011) Regulation of chromatin by histone modifications. *Cell Res.* 21, 381–395
72. Bartman, C.R. *et al.* (2019) Transcriptional burst initiation and polymerase pause release are key control points of transcriptional regulation. *Mol. Cell* 73, 519–532
73. Boeger, H. *et al.* (2008) Nucleosome retention and the stochastic nature of promoter chromatin remodeling for transcription. *Cell* 133, 716–726
74. Gilchrist, D.A. and Adelman, K. (2012) Coupling polymerase pausing and chromatin landscapes for precise regulation of transcription. *Biochim. Biophys. Acta* 1819, 700–706
75. Wu, S. *et al.* (2017) Independent regulation of gene expression level and noise by histone modifications. *PLoS Comput. Biol.* 13, e1005585
76. Muramoto, T. *et al.* (2010) Methylation of H3K4 Is required for inheritance of active transcriptional states. *Curr. Biol.* 20, 397–406
77. Vinuelas, J. *et al.* (2013) Quantifying the contribution of chromatin dynamics to stochastic gene expression reveals long, locus-dependent periods between transcriptional bursts. *BMC Biol.* 11, 15
78. Sood, V. *et al.* (2025) Identification of molecular determinants of gene-specific bursting patterns by high-throughput imaging screens. *Mol. Cell* 85, 913–928
79. Nicolas, D. *et al.* (2018) Modulation of transcriptional burst frequency by histone acetylation. *Proc. Natl. Acad. Sci. U. S. A.* 115, 7153–7158
80. Fraser, L.C.R. *et al.* (2021) Reduction in gene expression noise by targeted increase in accessibility at gene loci. *Proc. Natl. Acad. Sci. U. S. A.* 118, e2018640118
81. Ochiai, H. *et al.* (2020) Genome-wide kinetic properties of transcriptional bursting in mouse embryonic stem cells. *Sci. Adv.* 6, eaaz6699
82. Wu, Q. *et al.* (2023) Massively parallel characterization of CRISPR activator efficacy in human induced pluripotent stem cells and neurons. *Mol. Cell* 83, 1125–1139
83. Fritzsch, C. *et al.* (2018) Estrogen-dependent control and cell-to-cell variability of transcriptional bursting. *Mol. Syst. Biol.* 14, e7678
84. Gu, W. and Roeder, R.G. (1997) Activation of p53 sequence-specific DNA binding by acetylation of the p53 C-terminal domain. *Cell* 90, 595–606
85. Huang, Y. *et al.* (2021) An acetylation-enhanced interaction between transcription factor Sox2 and the steroid receptor coactivators facilitates Sox2 transcriptional activity and function. *J. Biol. Chem.* 297, 101389
86. Narita, T. *et al.* (2021) Enhancers are activated by p300/CBP activity-dependent PIC assembly, RNAPII recruitment, and pause release. *Mol. Cell* 81, 2166–2182
87. Welsh, S.A. and Gardini, A. (2023) Genomic regulation of transcription and RNA processing by the multitasking Integrator complex. *Nat. Rev. Mol. Cell Biol.* 24, 204–220
88. Baillat, D. *et al.* (2005) Integrator, a multiprotein mediator of small nuclear RNA processing, associates with the C-terminal repeat of RNA polymerase II. *Cell* 123, 265–276
89. Chuikov, S. *et al.* (2004) Regulation of p53 activity through lysine methylation. *Nature* 432, 353–360
90. Nasser, J. *et al.* (2021) Genome-wide enhancer maps link risk variants to disease genes. *Nature* 593, 238–243
91. Banerji, J. *et al.* (1981) Expression of a beta-globin gene is enhanced by remote SV40 DNA sequences. *Cell* 27, 299–308
92. Li, J. *et al.* (2019) Single-molecule nanoscopy elucidates RNA polymerase II transcription at single genes in live cells. *Cell* 178, 491–506
93. Boija, A. *et al.* (2018) Transcription factors activate genes through the phase-separation capacity of their activation domains. *Cell* 175, 1842–1855
94. Dufourt, J. *et al.* (2018) Temporal control of gene expression by the pioneer factor Zelda through transient interactions in hubs. *Nat. Commun.* 9, 5194
95. Levo, M. *et al.* (2022) Transcriptional coupling of distant regulatory genes in living embryos. *Nature* 605, 754–760
96. Chen, H. *et al.* (2018) Dynamic interplay between enhancer–promoter topology and gene activity. *Nat. Genet.* 50, 1296–1303
97. Zuin, J. *et al.* (2022) Nonlinear control of transcription through enhancer–promoter interactions. *Nature* 604, 571–577
98. Bartman, C.R. *et al.* (2016) Enhancer regulation of transcriptional bursting parameters revealed by forced chromatin looping. *Mol. Cell* 62, 237–247
99. Fukaya, T. *et al.* (2016) Enhancer control of transcriptional bursting. *Cell* 166, 358–368
100. Chen, L.F. *et al.* (2023) Structural elements promote architectural stripe formation and facilitate ultra-long-range gene regulation at a human disease locus. *Mol. Cell* 83, 1446–1461
101. Chen, L.F. *et al.* (2019) Enhancer histone acetylation modulates transcriptional bursting dynamics of neuronal activity-inducible genes. *Cell Rep.* 26, 1174–1188
102. Karr, J.P. *et al.* (2022) The transcription factor activity gradient (TAG) model: contemplating a contact-independent mechanism for enhancer–promoter communication. *Genes Dev.* 36, 7–16
103. Chen, Q. *et al.* (2023) Enhancer RNAs in transcriptional regulation: recent insights. *Front. Cell Dev. Biol.* 11, 1205540
104. Hamamoto, K. *et al.* (2023) Dynamic interplay between non-coding enhancer transcription and gene activity in development. *Nat. Commun.* 14, 826
105. Du, M. *et al.* (2024) Direct observation of a condensate effect on super-enhancer controlled gene bursting. *Cell* 187, 331–344
106. Cho, W.K. *et al.* (2018) Mediator and RNA polymerase II clusters associate in transcription-dependent condensates. *Science* 361, 412–415
107. Chong, S. *et al.* (2018) Imaging dynamic and selective low-complexity domain interactions that control gene transcription. *Science* 361, eaar2555
108. Sabari, B.R. *et al.* (2018) Coactivator condensation at super-enhancers links phase separation and gene control. *Science* 361, eaar3958
109. Bruckner, D.B. *et al.* (2023) Stochastic motion and transcriptional dynamics of pairs of distal DNA loci on a compacted chromosome. *Science* 380, 1357–1362
110. Gabriele, M. *et al.* (2022) Dynamics of CTCF- and cohesin-mediated chromatin looping revealed by live-cell imaging. *Science* 376, 496–501